

A background image of four students in a library. A young woman with long dark hair is on the left, smiling. Next to her is a young man with dark hair, also smiling. To his right is a young woman with glasses and dark hair, looking towards the right. On the far right is a young man with dark hair, seen from the back/side, looking at a laptop. They are all sitting at a table with books and a laptop. The background is filled with bookshelves.

Java 22

Unnamed Variables and Patterns

Background and Motivation

- Unnamed variables and unnamed patterns were introduced in Java 21 and finalised in Java 22.
- Both are denoted with the underscore character, `_`.
- The goal here is that in situations where variables are declared with no intention to use them, we can now make the intent of their *non-use* explicit.
- This makes the code cleaner, more readable and more maintainable.



Unnamed Variables

- Unnamed variables apply to local variables:
 - local variable declarations
 - headers in basic/enhanced *for* loops
 - try-with-resources
 - exception parameters in *catch* clauses
 - lambda parameters in lambda expressions.
- Unnamed variables do not apply to instance variables.
- Declaring a variable that is never used is very common in code whose side-effect is very important.



Unnamed Patterns

- Unnamed patterns enable us to ignore specific fields when using pattern matching.
- Particularly useful in record deconstruction and *switch* expressions.

